

# Onyx — GameKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **GameKit** (`SD:/lib/gamekit.so`, `user/Kits/gamekit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

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## What it is

GameKit is the games of the card's ROMs: the consoles the installed emulators play (their `app.txt`'s `games` =), the watched folders, the ROMs found in them by their extension, and their pictures -- what the Game Library and the console's home share.

|         |                                                                                                                         |
|---------|-------------------------------------------------------------------------------------------------------------------------|
|         |                                                                                                                         |
| Include | <code>#include "gamekit/gamekit.h"</code>                                                                               |
| Link    | <code>lib/gamekit.imp.a</code> (C++) or <code>lib/gamekit.imp_c.a</code> (C)                                            |
| Library | <code>SD:/lib/gamekit.so</code> — 9 entries in its table<br>( <code>user/Kits/gamekit/gamekit.abi</code> , append-only) |
| Sources | <code>user/Kits/gamekit/</code>                                                                                         |

## Using it

What the Game Library and the console's home share about the games: **the consoles** the installed emulators play (each emulator's `app.txt`: `games = Game Boy Color: gbc; Game Boy: gb, order = 50` — an emulator's package installed, its console appears), **the watched folders** (`SD:/apps/gamelib.app/config.ini`, one folder = line each; `SD:/roms` when none is said), **the ROMs** found in them and their sub-folders by their extension — sorted by console, then by name, each with a name to show (`ZeldaOracleOfSeason.gbc` → `Zelda Oracle Of Season`) — and **their pictures**, the title screens the Game Library makes (`SD:/apps/gamelib.app/thumbs/<file>.thm`, 160 × 144). The gamepads' buttons and their settings are not in it: `Include/gamepad.h` keeps them.

Every Game Boy Advance game, its picture (C):

```
#include "gamekit/gamekit.h"
```

```
static struct game_system sys[GAMES_SYSTEMS_MAX];  
static char dirs[GAMES_FOLDERS_MAX][GAMES_PATH];
```

```

static struct game g[256];
static unsigned pic[GAMES_THUMB_W * GAMES_THUMB_H];

int ns = games_systems (sys, GAMES_SYSTEMS_MAX);    // the consoles, by their
order
int nd = games_folders (dirs, GAMES_FOLDERS_MAX);    // the watched folders
int n = games_scan (dirs, nd, sys, ns, g, 256);      // by console, then by name
for (int i = 0; i < n; i++)
    if (ax_streq (sys[g[i].sys].name, "Game Boy Advance"))
    {
        ax_putln (g[i].name);                        // "Star Courier"
        if (games_thumb_load (&g[i], pic)) { /* 0xRRGGBB, 160 x 144 */ }
    }

```

A game is opened with its emulator by AppKit (`lx_open (g[i].path, "")`: the emulator found by the same `app.txt`). A folder added: `games_folders_save (dirs, nd + 1)`. A program that makes the pictures saves them with `games_thumb_save`.

## Index

Everything the headers declare, in their order — the details are in each header's part below.

| Name                            | What it does                                                                                                                  | Header                 |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <code>game_system</code>        | (a type)                                                                                                                      | <code>gamekit.h</code> |
| <code>game</code>               | (a type)                                                                                                                      | <code>gamekit.h</code> |
| <code>games_systems</code>      | The consoles of the installed emulators (SD:/apps/*.app/app.txt with a <code>games</code> line)                               | <code>gamekit.h</code> |
| <code>games_folders</code>      | The watched folders (GAMES_CONFIG)                                                                                            | <code>gamekit.h</code> |
| <code>games_folders_save</code> | The watched folders written (GAMES_CONFIG)                                                                                    | <code>gamekit.h</code> |
| <code>games_system_of</code>    | The console of a file by its extension → its index in <code>sys</code> , -1 none.                                             | <code>gamekit.h</code> |
| <code>games_nice_name</code>    | A file's name to show                                                                                                         | <code>gamekit.h</code> |
| <code>games_scan</code>         | The ROMs of the folders (6 sub-folders deep at most), sorted by console ( <code>sys</code> 's order) then by name → how many. | <code>gamekit.h</code> |
| <code>games_thumb_path</code>   | A game's picture                                                                                                              | <code>gamekit.h</code> |
| <code>games_thumb_load</code>   | A game's picture                                                                                                              | <code>gamekit.h</code> |
| <code>games_thumb_save</code>   | A game's picture                                                                                                              | <code>gamekit.h</code> |

## gamekit/gamekit.h

gamekit.h -- GameKit (SD:/lib/gamekit.so; docs/06-KITS-GUIDE.md "GameKit", docs/20-GAMEKIT.md): the games of the card's ROMs: the consoles the installed emulators play (each emulator's app.txt: games = Game Boy Color: gbc; Game Boy: gb, order = 50), the watched folders (the Game Library's SD:/apps/gamelib.app/config.ini, one folder = SD1:/Roms line each; SD:/roms by default), the ROMs found in them (and their sub-folders) by their extension, and their pictures (the title screens the Game Library makes, cached in SD:/apps/gamelib.app/thumbs/.thm: 160 × 144 RGB). The Game Library and the console's shell read the same catalogue through it. C and C++: link lib/gamekit.imp.a (C++) or lib/gamekit.imp\_c.a (C); the interface is append-only (gamekit.abi). (The gamepads' buttons and their settings stay in Include/gamepad.h.)

```
struct game_system sys[GAMES_SYSTEMS_MAX]; int ns = games_systems (sys,
GAMES_SYSTEMS_MAX);
char dirs[GAMES_FOLDERS_MAX][GAMES_PATH]; int nf = games_folders (dirs,
GAMES_FOLDERS_MAX);
static struct game g[256]; int n = games_scan (dirs, nf, sys, ns, g, 256); // by
console, then by name
unsigned *px = new unsigned[GAMES_THUMB_W * GAMES_THUMB_H]; if (games_thumb_load
(&g[0], px)) ...
```

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```
#define GAMES_SYSTEMS_MAX 12
#define GAMES_FOLDERS_MAX 8
#define GAMES_PATH 200
#define GAMES_CONFIG "SD:/apps/gamelib.app/config.ini" // the watched
folders
#define GAMES_THUMBS "SD:/apps/gamelib.app/thumbs/" // the pictures:
<file>.thm
#define GAMES_THUMB_W 160
#define GAMES_THUMB_H 144

struct game_system // a console an emulator plays
{
    char name[40]; // "Game Boy Advance"
    char ext[64]; // its files' extensions, " gba " (lower case, each
between spaces)
    char ext_text[64]; // the same to show: ".gba" (".sfc / .smc")
    char emu[24]; // its emulator: SD:/apps/<emu>.app ("gbaemu")
    int order; // the consoles' order (app.txt's order x 16 + its rank in
'games'): small first
};
struct game // a ROM
{
    char path[GAMES_PATH]; // "SD1:/Roms/gba/Star_Courier.gba"
    char name[64]; // its name to show: "Star Courier"
    char key[64]; // its file's name (its picture's: GAMES_THUMBS + key +
".thm")
    int sys; // its console: an index in the game_system array it was
scanned with
};
```

The consoles of the installed emulators (SD:/apps/\*.app/app.txt with a `games` line; the emulators of before that key by their names), sorted by order → how many (at most max).

```
int games_systems (struct game_system *out, int max);
```

The watched folders (GAMES\_CONFIG; none said: SD:/roms; `folder` = with no value: none) → how many.

```
int games_folders (char (*out)[GAMES_PATH], int max);
```

The watched folders written (GAMES\_CONFIG: one `folder` = line each; n 0: none, not the default) → 1 written.

```
int games_folders_save (const char (*folders)[GAMES_PATH], int n);
```

The console of a file by its extension → its index in sys, -1 none.

```
int games_system_of (const char *file, const struct game_system *sys, int nsys);
```

A file's name to show: "ZeldaOracleOfSeason.gbc" → "Zelda Oracle Of Season", "super\_mario" → "Super mario".

```
void games_nice_name (const char *file, char *out, int cap);
```

The ROMs of the folders (6 sub-folders deep at most), sorted by console (sys's order) then by name → how many.

```
int games_scan (const char (*folders)[GAMES_PATH], int nfolders, const struct game_system *sys, int nsys, struct game *out, int max);
```

A game's picture: its file's path (GAMES\_THUMBS + key + ".thm"); read into px (GAMES\_THUMB\_W x GAMES\_THUMB\_H, 0xRRGGBB) → 1, 0 none made yet; written from px → 1 written.

```
void games_thumb_path (const struct game *g, char *out, int cap);  
int games_thumb_load (const struct game *g, unsigned *px);  
int games_thumb_save (const struct game *g, const unsigned *px);
```

```
}  
#endif
```